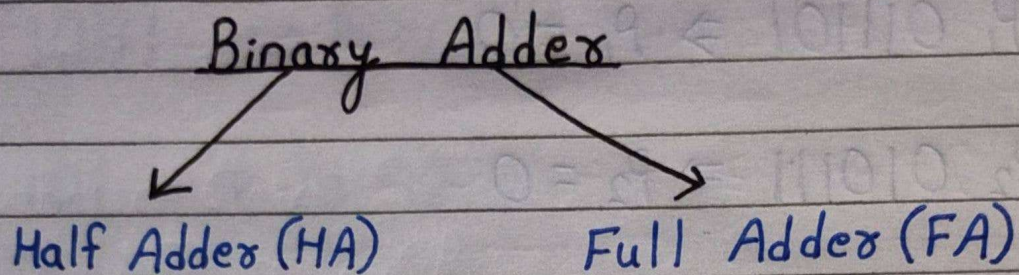


Module - II

Combinational Logic

Combinational Logic Circuit



Half Adder

Truth Table:

	A	B	Sum (S)	Carry (C)
0	0	0	0	0
1	0	1	1	0
2	1	0	1	0
3	1	1	0	1

K-map:

A \ B	$\bar{B}$	B
$\bar{A}$		1
A	1	

A \ B	$\bar{B}$	B
$\bar{A}$	0	1
A	2	1

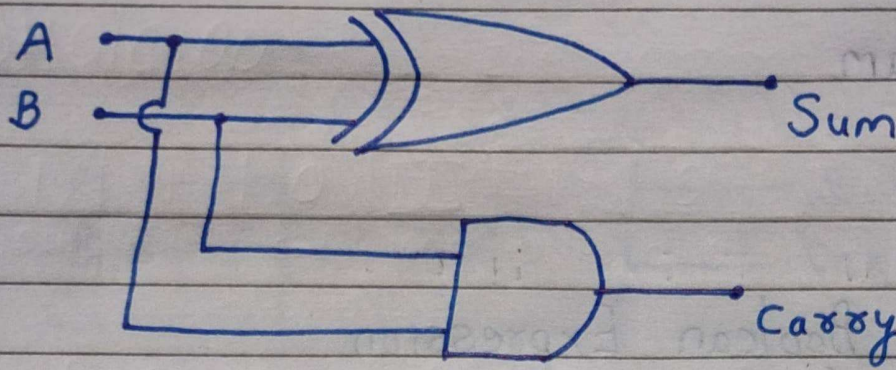
$$\text{Sum} = \bar{A}B + A\bar{B}$$

$$\text{Carry} = AB$$

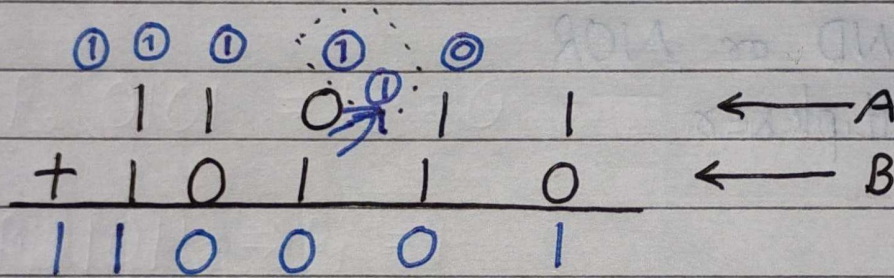
$$\Rightarrow \text{Sum} = A \oplus B \text{ [XOR]}$$

$$\Rightarrow \text{Carry} = A \cdot B \text{ [AND]}$$

Logic Circuit:

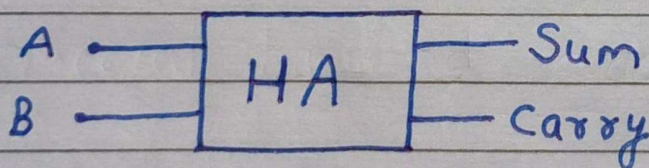


Example:



Here, if a carry is obtained then another half adder is required, which is the limitation.

Block Diagram:



Design of Any Logic Circuit [Steps]

1. Block diagram

2. Truth table

3. K-map for Boolean Expression

4. Design

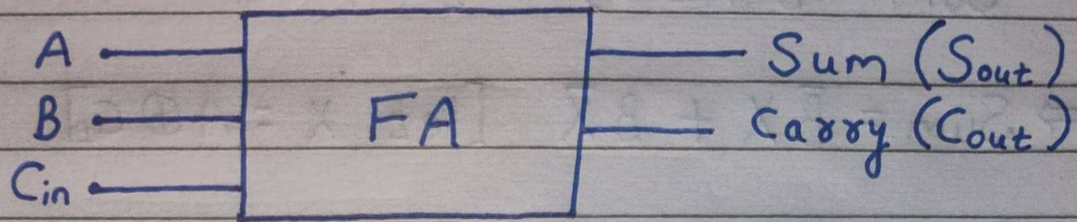
→ Basic Gate

→ NAND or NOR

→ Multiplexer

Full Adder

Block Diagram:



Truth Table:

	A	B	C _{in}	Sum	Carry
0	0	0	0	0	0
1	0	0	1	1	0
2	0	1	0	1	0
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	1
6	1	1	0	0	1
7	1	1	1	1	1

K-map:

A \ B C _{in}	$\bar{B}\bar{C}_{in}$	$\bar{B}C_{in}$	BC_{in}	$B\bar{C}_{in}$
$\bar{A}$				
A				

← For Sum

$$\text{Sum} = A\bar{B}\bar{C}_{in} + \bar{A}\bar{B}C_{in} + ABC_{in} + \bar{A}B\bar{C}_{in}$$

$$\Rightarrow \text{Sum} = \bar{B}(A\bar{C}_{in} + \bar{A}C_{in}) + B(AC_{in} + \bar{A}\bar{C}_{in})$$

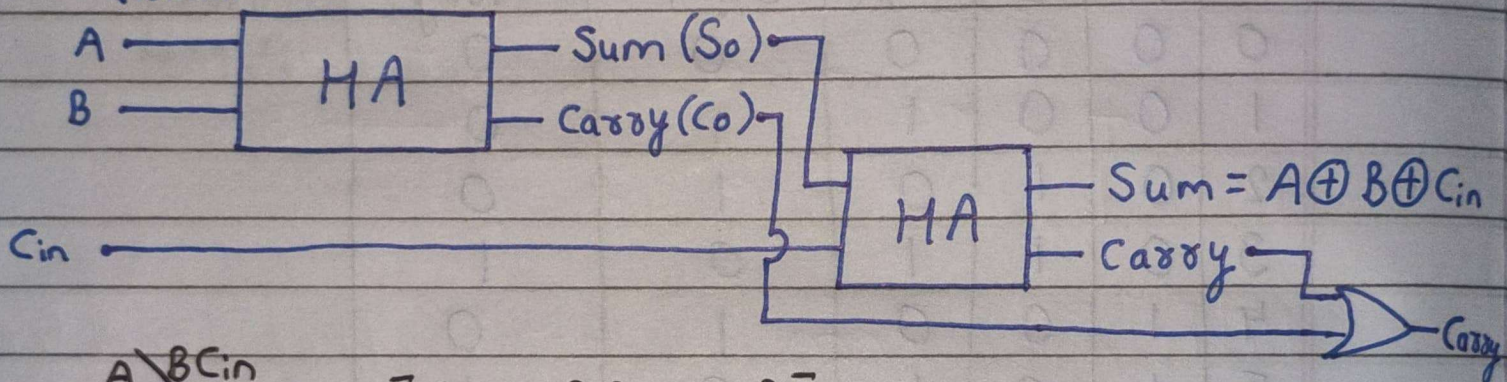
$$\Rightarrow \text{Sum} = \bar{B}(\underset{\substack{\uparrow \\ \text{XOR}}}{A \oplus C_{in}}) + B(\underset{\substack{\uparrow \\ \text{X-NOR}}}{A \odot C_{in}})$$

$$\Rightarrow \text{Sum} = \bar{B}(A \oplus C_{in}) + B(\overline{A \oplus C_{in}})$$

$$\Rightarrow \text{Sum} = \bar{B}X + B\bar{X} \quad [\text{let } X = A \oplus C_{in}]$$

$$\Rightarrow \text{Sum} = B \oplus X$$

$$\Rightarrow \underset{\substack{\text{(Sout)}}}{\text{Sum}} = A \oplus B \oplus C_{in}$$

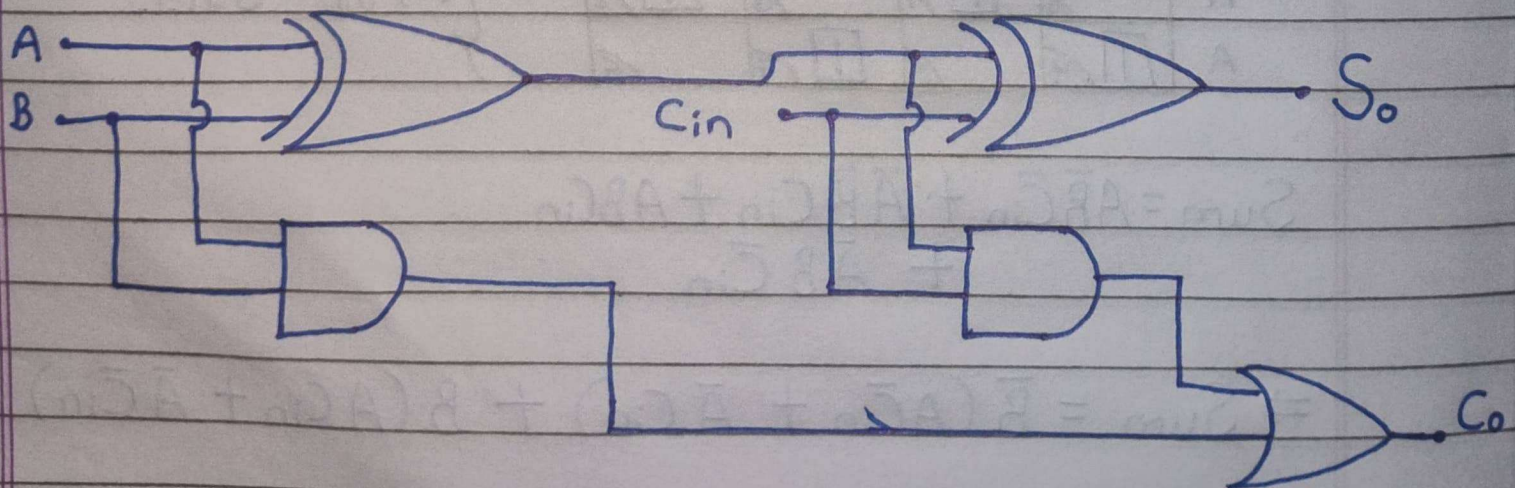


A \ BC _{in}	$\bar{B}\bar{C}_{in}$	$\bar{B}C_{in}$	BC_{in}	$B\bar{C}_{in}$	
$\bar{A}$	0	1	1	3	2
A	4	1	1	7	6

} ← For Carry

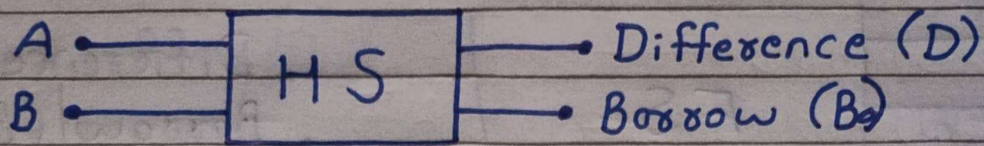
$$\underset{\substack{\text{(Cout)}}}{\text{Carry}} = BC_{in} + AC_{in} + AB$$

Logic Circuit:



Half Subtractor

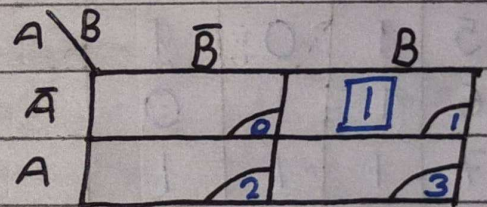
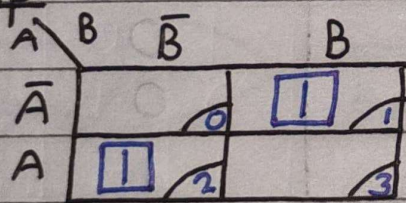
Block Diagram:



Truth Table:

	A	B	Difference (D)	Borrow (B ₀)
0	0	0	0	0
1	0	1	1	1
2	1	0	1	0
3	1	1	0	0

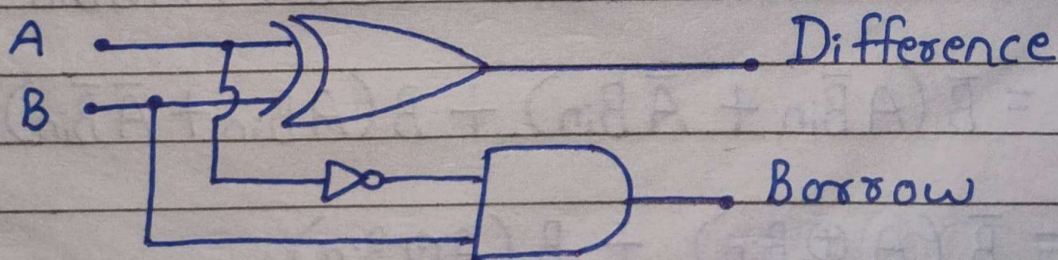
K-map:



$$\begin{aligned}\text{Difference} &= A\bar{B} + \bar{A}B \\ &= A \oplus B \\ &\quad [\text{XOR}]\end{aligned}$$

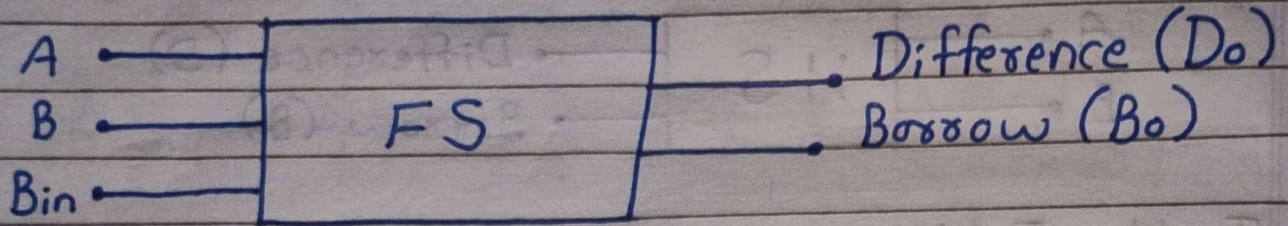
$$\text{Borrow} = \bar{A}B$$

Logic Circuit:



Full Subtractor

Block Diagram:



Truth Table:

	A	B	Bin	Difference (Do)	Borrow (Bo)
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	1
4	1	0	0	1	0
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

K-map:

	$B\bar{B}in$	$\bar{B}\bar{B}in$	$\bar{B}Bin$	$B\bar{B}in$	$B\bar{B}in$
$\bar{A}$	0	1		3	2
A	4		5	7	6

← For D_0

$$\text{Difference } (D_0) = A\bar{B}\bar{B}in + \bar{A}\bar{B}Bin + AB\bar{B}in + \bar{A}B\bar{B}in$$

$$\Rightarrow D_0 = \bar{B}(A\bar{B}in + \bar{A}Bin) + B(A\bar{B}in + \bar{A}\bar{B}in)$$

$$\Rightarrow D_0 = \bar{B}(A \oplus Bin) + B(A \odot Bin)$$

$$\Rightarrow D_0 = \bar{B} (A \oplus B_{in}) + B (\overline{A \oplus B_{in}})$$

$$\Rightarrow D_0 = \bar{B} X + B \bar{X} \quad [\text{Let } X = A \oplus B_{in}]$$

$$\Rightarrow D_0 = B \oplus X$$

$$\Rightarrow \text{Difference } (D_0) = A \oplus B \oplus B_{in}$$

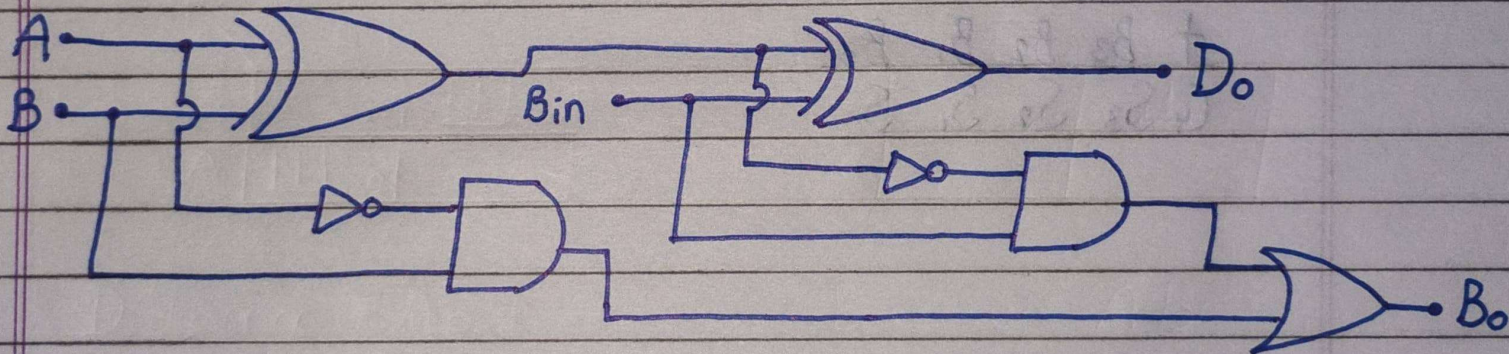
A \ B B _{in}	$\bar{B} \bar{B}_{in}$	$\bar{B} B_{in}$	$B B_{in}$	$B \bar{B}_{in}$	
$\bar{A}$	0	1	1	1	2
A	4	5	1	7	6

← For B₀

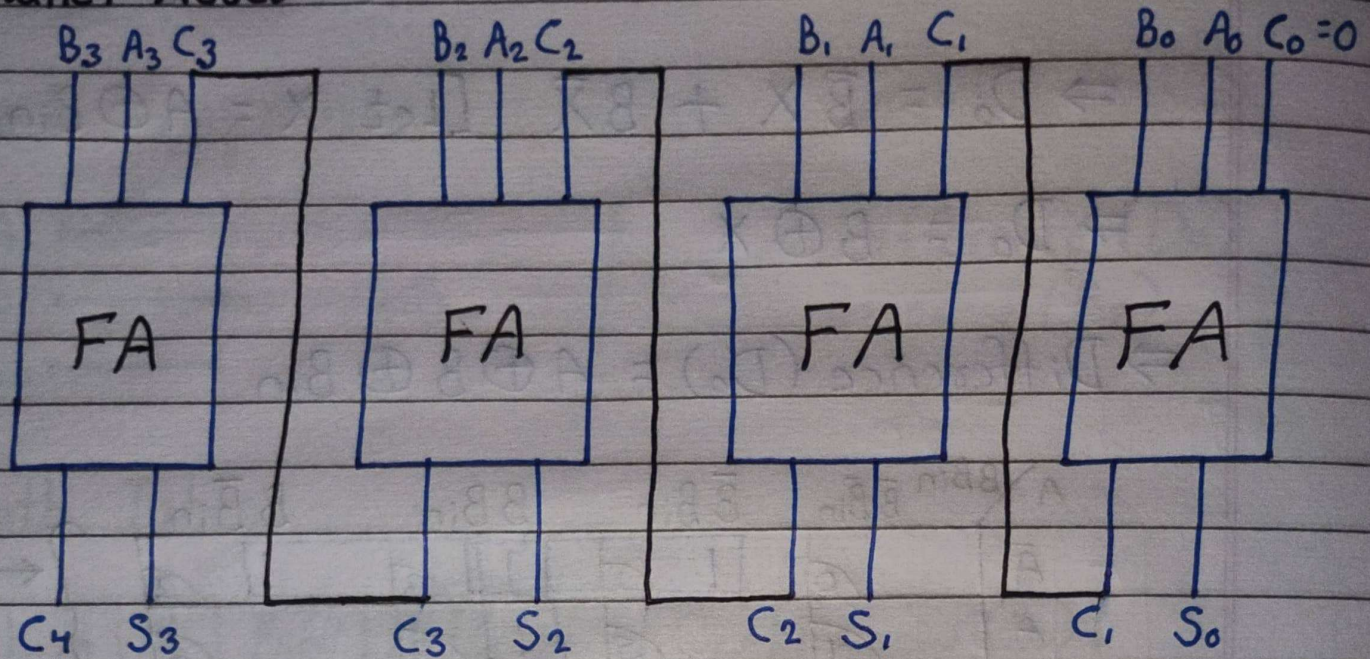
$$\text{Borrow } (B_0) = \bar{A} B_{in} + B B_{in} + \bar{A} B$$

$$\text{or } B_0 = \bar{A} B + B_{in} (\bar{A} + B)$$

Logic Circuit:



Parallel Adder



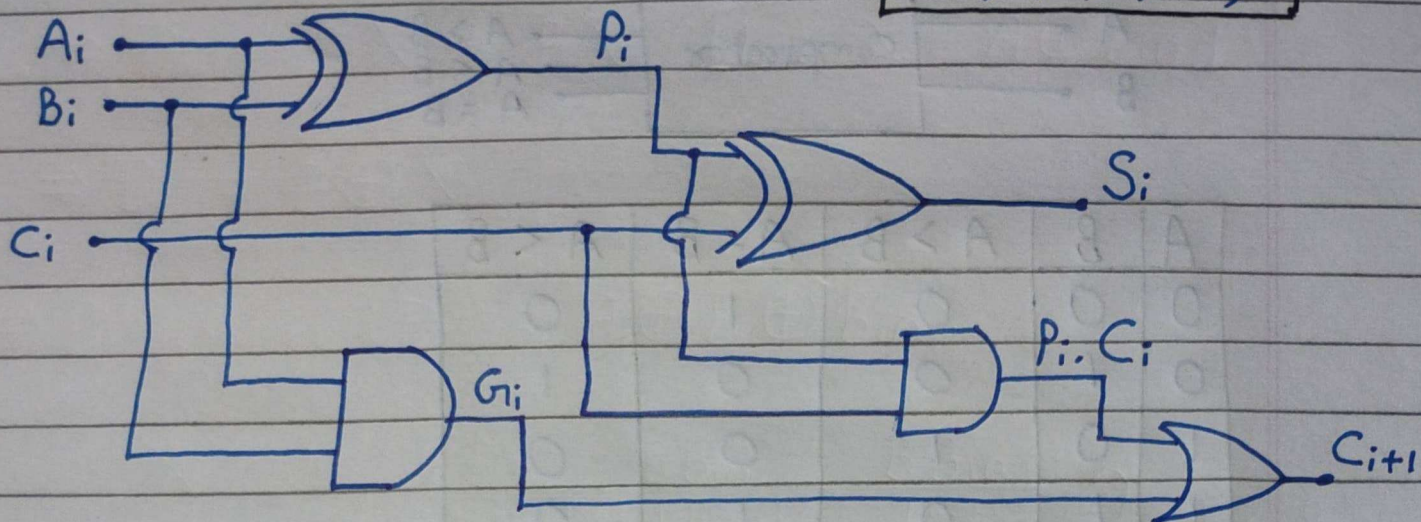
Final Sum:

C_4	S_3	S_2	S_1	S_0
-------	-------	-------	-------	-------

$$\begin{array}{r}
 C_4 \ C_3 \ C_2 \ C_1 \ C_0 \\
 A_3 \ A_2 \ A_1 \ A_0 \\
 + \ B_3 \ B_2 \ B_1 \ B_0 \\
 \hline
 C_4 \ S_3 \ S_2 \ S_1 \ S_0
 \end{array}$$

Look Ahead Carry Adder

$$i = 0, 1, 2, 3, \dots, n$$



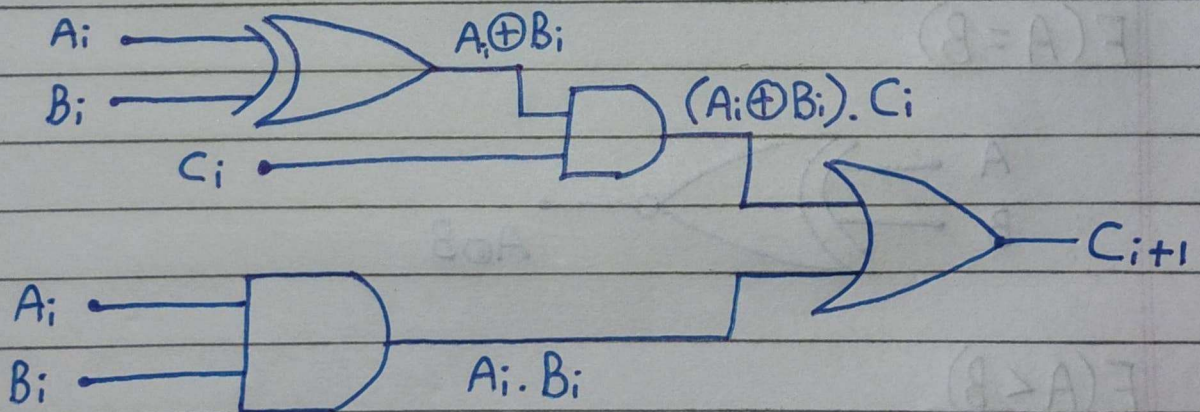
$$P_i = A_i \oplus B_i$$

$$G_i = A_i \cdot B_i$$

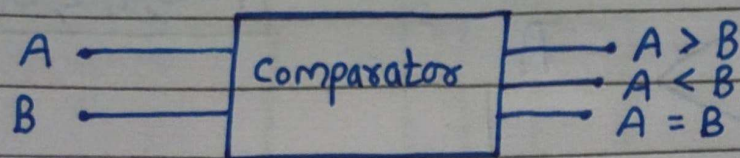
$$S_i = P_i \oplus C_i = A_i \oplus B_i \oplus C_i$$

$$C_{i+1} = G_i + P_i \cdot C_i$$

$$\Rightarrow C_{i+1} = A_i \cdot B_i + (A_i \oplus B_i) \cdot C_i$$



Comparators



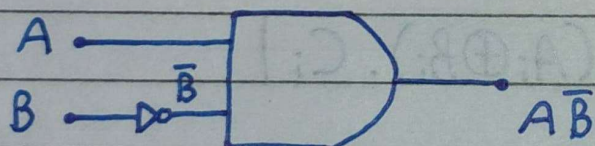
A	B	$A > B$	$A = B$	$A < B$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

$$F(A > B) = A\bar{B}$$

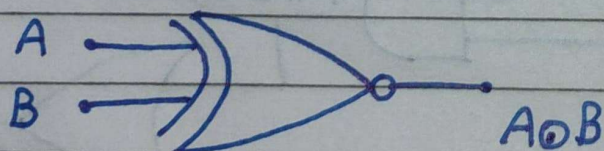
$$F(A = B) = \bar{A}\bar{B} + AB = A \odot B$$

$$F(A < B) = \bar{A}B$$

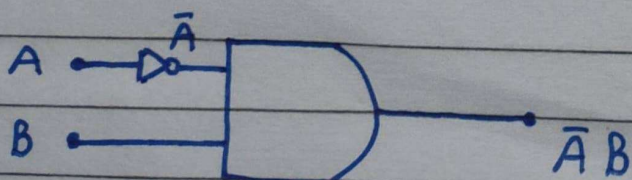
$$F(A > B)$$



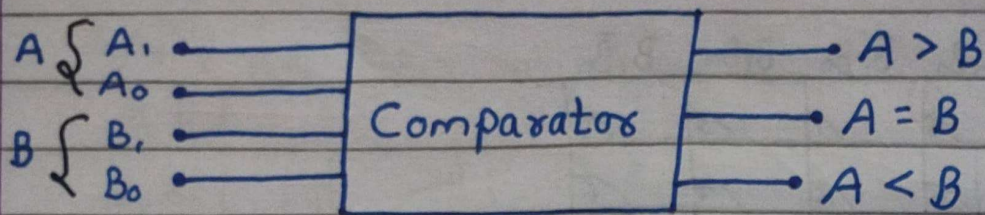
$$F(A = B)$$



$$F(A < B)$$



Comparators [2 Bit]



	A		B		A > B	A = B	A < B
	A ₁	A ₀	B ₁	B ₀			
0	0	0	0	0	0	1	0
1	0	0	0	1	0	0	1
2	0	0	1	0	0	0	1
3	0	0	1	1	0	0	1
4	0	1	0	0	1	0	0
5	0	1	0	1	0	1	0
6	0	1	1	0	0	0	1
7	0	1	1	1	0	0	1
8	1	0	0	0	1	0	0
9	1	0	0	1	1	0	0
10	1	0	1	0	0	1	0
11	1	0	1	1	0	0	1
12	1	1	0	0	1	0	0
13	1	1	0	1	1	0	0
14	1	1	1	0	1	0	0
15	1	1	1	1	0	1	0

K-map for $A > B$

$A, A_0 \backslash B, B_0$	$\bar{B}, \bar{B}_0$	$\bar{B}, B_0$	B, B_0	$B, \bar{B}_0$
$\bar{A}, \bar{A}_0$	0	1	3	2
$\bar{A}, A_0$	4	5	7	6
A, A_0	12	13	15	14
$A, \bar{A}_0$	8	9	11	10

$$F(A > B) = A, \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + \cancel{A_0 \bar{B}_1 \bar{B}_0} + A, A_0 \bar{B}_0$$

K-map for $A = B$

$A, A_0 \backslash B, B_0$	$\bar{B}, \bar{B}_0$	$\bar{B}, B_0$	B, B_0	$B, \bar{B}_0$
$\bar{A}, \bar{A}_0$	1	0	3	2
$\bar{A}, A_0$	4	5	7	6
A, A_0	12	13	15	14
$A, \bar{A}_0$	8	9	11	10

$$F(A = B) = \bar{A}, \bar{A}_0 \bar{B}_1 \bar{B}_0 + \bar{A}, A_0 \bar{B}_1 B_0 + A, A_0 B_1 B_0 + A, \bar{A}_0 B_1 \bar{B}_0$$

$$= \bar{A}, \bar{B}_1 (\underbrace{\bar{A}_0 \bar{B}_0 + A_0 B_0}_{X-NOR}) + A, B_1 (\underbrace{A_0 B_0 + \bar{A}_0 \bar{B}_0}_{X-NOR})$$

$$= (A_0 \odot B_0) (\underbrace{\bar{A}, \bar{B}_1 + A, B_1}_{X-NOR})$$

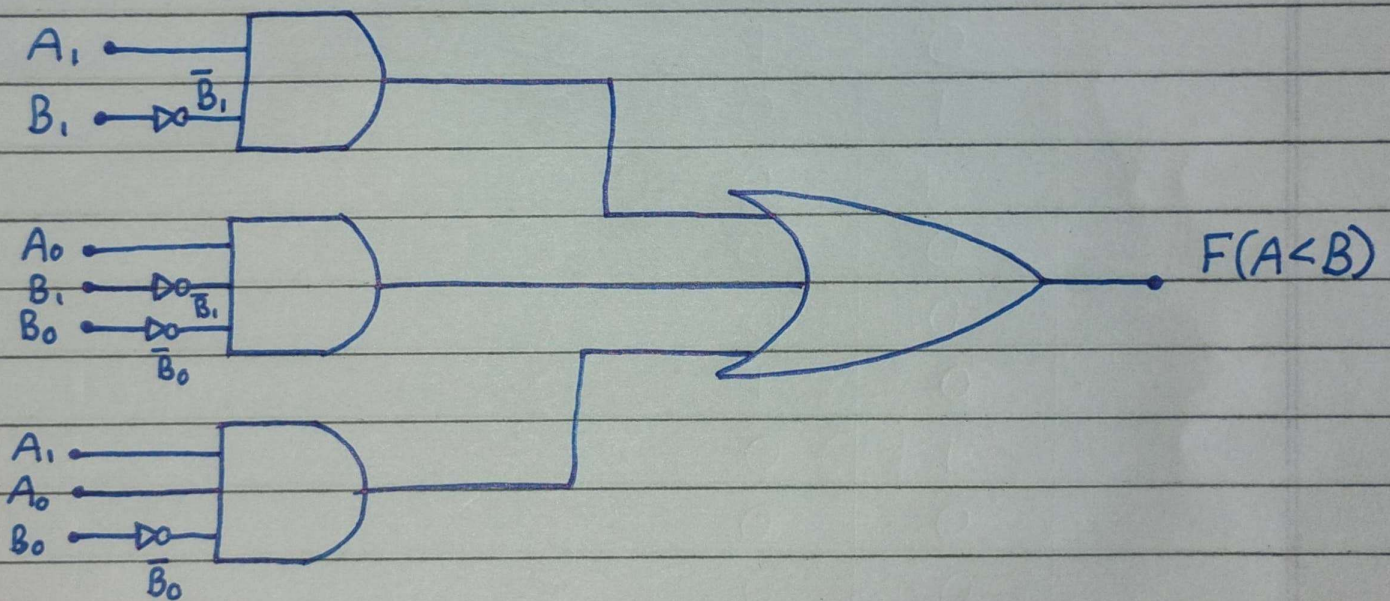
$$= (A_0 \odot B_0) (A_1 \odot B_1)$$

K-map for $A < B$

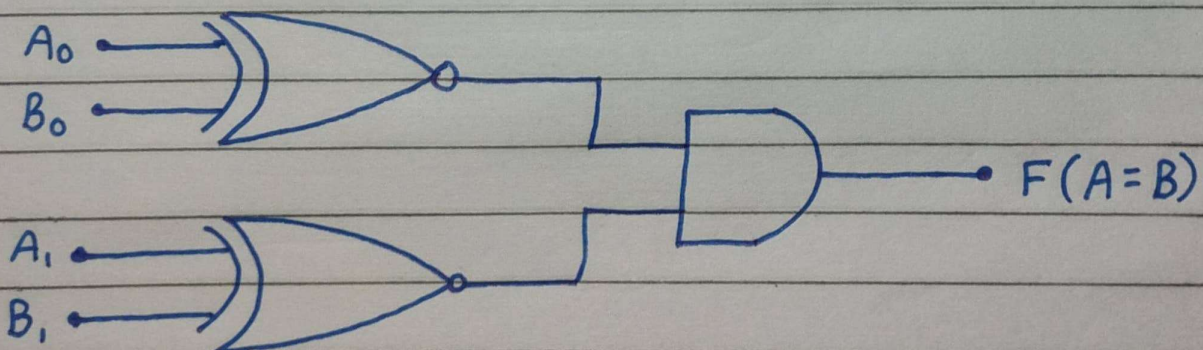
$A, A_0 \backslash B, B_0$	$\bar{B}, \bar{B}_0$	$\bar{B}, B_0$	B, B_0	$B, \bar{B}_0$
$\bar{A}, \bar{A}_0$	0	1	1	2
$\bar{A}, A_0$	4	5	1	6
A, A_0	12	13	15	14
$A, \bar{A}_0$	8	9	11	10

$$F(A < B) = \bar{A}_1 \bar{B}_1 + \bar{A}_1 \bar{A}_0 B_0 + \bar{A}_0 B_1 B_0$$

$A > B$



$A = B$



$A < B$

